

Multi-modal perception

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To design, optimise and deliver multimedia and virtual-reality products and services it is necessary to match performance to the capabilities of users. When a multimedia system is used, the presence of audio and video stimuli introduces significant cross-modal effects (the sensory streams interact). This paper introduces a number of cross-modal interactions that are relevant to communications systems and discusses the advanced experimental techniques required to provide data for modelling multi-modal perception. The aim of the work is to provide a multi-modal perceptual model that can be used for performance assessment and can be incorporated into coding algorithms. The current and future applications of multi-modal modelling are discussed.

1. Introduction

Improvements in available data transmission and storage bandwidth, market deregulation, and the convergence between communications, computing, and content, are all acting to accelerate the introduction of new and advanced multimedia and telepresence products and services. BT has made experimental trials of advanced shared spaces [1, 2], investigated the application of telepresence to medicine [3] and education [4], and is researching the emerging interface technologies for interactive collaborative media-environments [5]. In addition to diversification in products and services, network and delivery technology is also evolving rapidly. In particular it is increasingly common for both national and international traffic to encounter nonlinear processing such as data reduction, echo-cancellation or delay variation due to packet-based transport layers. To provide and maintain required levels of customer satisfaction in an efficient manner, it is desirable to have objective measures of quality that relate to perceived performance. Models of hearing have been developed to predict subjective speech and audio quality [6, 7]. Models of visual perception are also in development [8, 9]. For multimedia and virtual-reality systems, multisensory models will be required. Research is now being directed towards developing a multi-modal model that can be used to predict audio-video quality. The goal of the multi-modal model is to fulfil a similar role for audio-video systems to that which the audio model currently performs for audio systems.

Understanding and predicting multi-modal performance is particularly difficult when a task situation is dynamically varying — which is not unlikely. Sensory interactions, and the attention given to particular aspects of performance, are dependent on the nature of the task undertaken. It is anticipated that to understand and eventually model multi-sensory perception it will be necessary to understand and

model the underlying cognitive processes. The use of knowledge about the underlying cognitive mechanisms to inform model structure has two advantages. Firstly, the model is made more representative of actual human quality perception. Secondly, the critical sensory and perceptual components of the model can be identified and weighted according to the principles of human information-processing.

Current experimental investigations are providing knowledge of multi-modal perception to allow the development of multisensory perceptual models. This knowledge is also of immediate use to designers/technologists who require insight into the gross cross-modal behaviours relevant to improving perceived performance.

This paper introduces the main aspects of multi-modal perception and recent experimental investigations undertaken at BT Laboratories (BTL) to provide the data required for the development of a multi-sensory model. The subjective test methods traditionally used in the communications industry have had to be supplemented with a variety of new techniques and these are also described together with future trends in subjective testing. Finally the current and future application of multi-modal models are discussed.

1.1 Multi-modal perceptual models

BT Laboratories already have considerable experience in the development of auditory perceptual models. A practical telephony bandwidth model is used by BT for quality assessments of equipment both in development and in the field. A visual perceptual model is also under development, based on a similar architecture. A description of auditory and visual perceptual modelling is given in Hollier and Voelcker [10] together with a description of

example applications. In addition, work is under way to combine these two models to produce a multi-modal model which will provide similar information for systems which utilise both audio and video. Inclusion of the other senses is expected to follow in due course.

The structure of the audio perceptual model is in two layers — the sensory layer (reproducing the large-scale properties of hearing) and the perceptual layer (which attempts to predict the subjective importance of audible errors). The perceptual layer approximates the higher level perceptual processes that determine the subjectivity of errors and leads to the formation of an opinion.

The visual model has a similar architecture to the auditory model with a sensory and perceptual layer. The perceptual layer is somewhat more complex for the visual model, predicting the subjective importance of visible errors by relating them to perceptually relevant image components.

The multi-modal model combines these unisensory models using the perceptually weighted signals from the auditory and visual models as the basis for multi-modal predictions. The multi-modal model must account for cross-modal interactions as well as task influences in order to give a task-related perceived performance metric. This does not infer that task has no bearing on user perceptions in monomodal cases, such as an audio-only telephony call. Current mono-modal models are designed to work with

generic cases such as simple information exchange. Multi-modal applications, by their very nature, are more complex and varied and hence the importance of task increases. In addition, interactions between the perceptions of the different modalities may also be confounded with task, further underlining the importance of understanding task in relation to multi-modal perception. A block diagram of the proposed model architecture is given in Fig 1.

It is important to note that the auditory and visual models produce a number of parameters describing the subjectivity of audible and visible errors respectively. It has previously been shown [11] that such error descriptors can be designed to map to different subjective opinion scales. This provides evidence that low-level parameters can be found that can be weighted to predict overall opinions that are biased by different task demands.

2. Cognitive processes, memory and task

The key difference between mono-modal and multi-modal models is the need to explicitly account for aspects of the underlying cognitive behaviour, and the influence of the task undertaken. Aspects of cognition, memory and task dependency are discussed in this section.

2.1 Cognitive processes and quality evaluations

Psychological models of human decision-making tend to concentrate on identifying those aspects of human

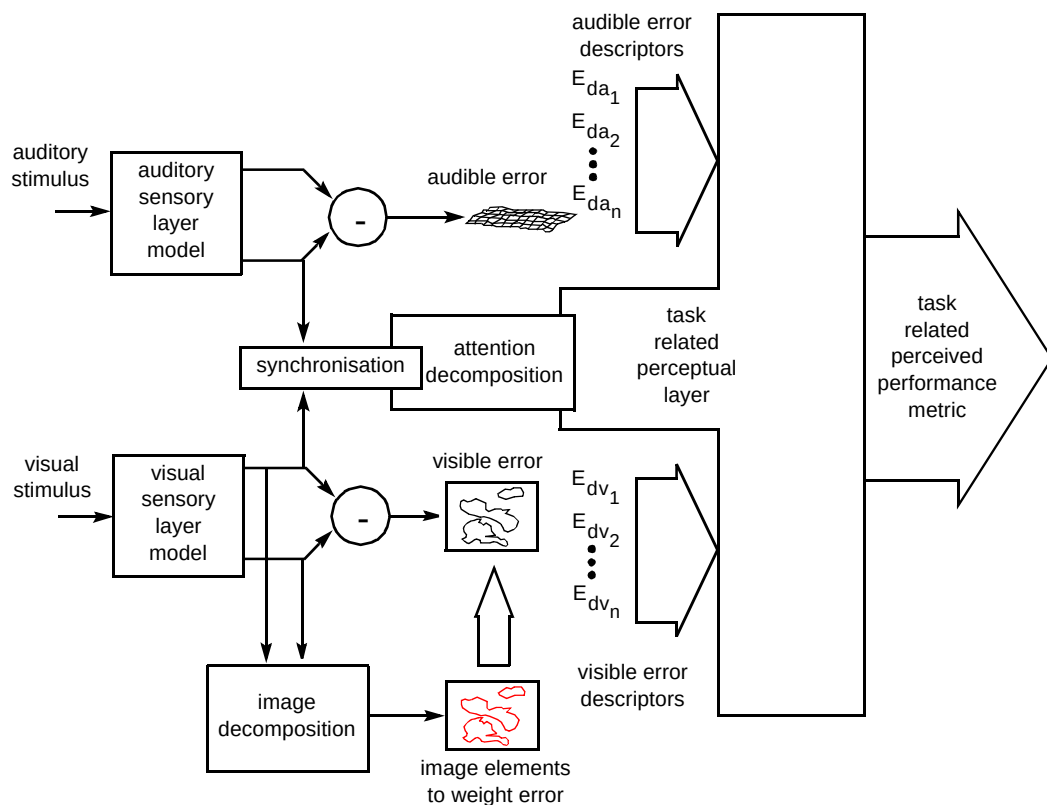


Fig 1 Proposed architecture for multisensory perceptual model [10].

information-processing responsible for specific human behaviours. For example, jury decision-making processes are heavily influenced by the perceived personal attributes of a witness often at the expense of the testimony produced by the witness. The focus of the present research is on understanding human perception and evaluation of the audio-video quality for some communications system. The psychophysical components of the multi-modal model have already been introduced. Although the individual sensory models (and the proposed architecture for the multi-modal model) do not explicitly model the cognitive processes which lead to an individual's judgement of quality, knowledge and understanding of these processes is of crucial importance to the success of the model. BT is at the forefront of research into understanding subjective reactions to audio-video quality because its research takes account of how sensory, perceptual and cognitive processes act on quality evaluations. The cognitive processes underlying decision-making are introduced below and discussed in the context of subjective quality evaluation.

Cognition is a field of psychology concerned with understanding the mental processes involved in thinking, problem-solving, memory, learning, perception and decision-making. Memory, perceptual and decision-making processes appear most critical to understanding human evaluation of audio-video quality. The perceptual processing involved in subjective quality assessments has already been introduced. The present discussion will focus on decision-making processes which can affect quality evaluation and will examine the most influential structures, such as human memory, which are called upon by the information-processing mechanisms active in human decision-making.

2.2 Decision-making processes

Consider the typical experimental situation within which subjective quality testing is performed. Subjects view a series of test clips; at the end of each clip an evaluation of the overall audio-video quality for the clip is made. Clips are maybe 5 or 10 seconds in duration and the content and quality may be inconsistent during that time period. Audio errors are qualitatively different from video errors.

Now consider a simplified version of the subjects' task. For each clip, the subject has to consider the quality of the audio and that of the video at each moment in time for the full clip. The changes in audio quality during the clip as well as changes in video quality have to be detected and given some weighting towards the final quality evaluation. The evaluation is a single, retrospective event. Even choosing to ignore such components as the past knowledge and experience of subjects (e.g. they may have poor reception quality at home, or be especially interested in the content of the clip), the decision-making task of the subject

is complex. The errors have both spatial and temporal properties — the intensity, duration, and frequency of errors can change during a single clip. Most important of all, errors occurring in the audio and video must be simultaneously processed and integrated into the evaluation. How is this achieved? What properties of the stimulus are most important to the decision-making operations?

Subjective quality evaluation is dependent on a number of factors, for example the sensory and perceptual ability of subjects to identify errors in the audio and video signals presented. Once error information has been extracted, it has to be collected together, or integrated, into some working decision-making process within which new information updates the existing evaluation. Previous experimental work on human decision-making has identified four critical cognitive components acting on judgements and evaluations [12]:

- long-term memory,
- working memory,
- attentional processes,
- current goal or task.

In the current framework of quality evaluation, the subjects' goal is to arrive at some decision of quality for the system under test. Therefore, attentional processes are directed towards selecting out information relevant to the task. Our ability to selectively attend to external environmental cues is dependent both on the properties of the stimulus and the demands acting on the human information-processing resources. The latter point is especially important, as audio (or video) errors may not be noticed if errors are occurring in both modalities. This may be characterised as a form of cross-modal error masking where attention is grabbed by errors in one modality to such an extent that errors in the other modality are either neglected or not registered. Information which has been perceived and identified as important to the current goal is then integrated into the decision-making process.

When information has been gathered together, other cognitive factors operate to produce the final evaluation. Long-term memory stores information that has been learned or experienced in the past. Information present in the experiment may activate long-term memory associations and this information will have some influence on a subject's formation of overall quality ratings. Working memory is a short-term memory system used for temporarily storing and manipulating information [13]. Working memory consists of three components — central executive, phonological loop, and visuo-spatial sketch pad. Verbal information is processed by the phonological loop and visual-spatial

information by the visuo-spatial sketch pad. The central executive controls the allocation of resources within working memory, and is responsible for combining information present in the phonological loop and the visuo-spatial sketch pad. Information stored in working memory can be passed into long-term memory using such strategies as rehearsal. The precise role of these memory processes in quality evaluations is unknown as yet, but research has reported consistent working memory influences on quality evaluations. For example, Aldridge et al [14] found that subjects give poorer ratings of video quality when the worst quality video appeared at the end compared to the beginning of a 30 sec clip (the so-called recency effect).

One of the properties of nonlinear data processing of audio-video material is that errors can vary in intensity, duration, and frequency. All three properties have the potential to affect subjective evaluations of quality. Previous research has shown that evaluations are more heavily influenced by the intensity of an event, and that the duration of that event is relatively unimportant [15, 16]. For example, it has been reported that retrospective evaluations of video quality are sensitive to the peak intensity (i.e. the worst part of the picture) and that quality ratings were the same whether the poor quality video was 5 sec or 10 sec in duration [17]. The decision-making process is therefore more attuned to identifying peaks in error intensity than to the time-course of an error. Work on frequency of errors has not been reported and this factor clearly requires attention. Similarly, an extension of the work on intensity and duration to encompass audio-visual stimuli would be of much use in developing a multi-modal model of subjective quality perception.

Research into understanding the cognitive processes operating on subjective quality evaluations is complementary to the sensory and perceptual modelling work described previously. Information is first filtered by the sensory and perceptual auditory-visual mechanisms. The decision-making process then acts upon the information which has been selected from the audio-visual array. The role of the decision-making operator is to determine the relevance of the available information to the working hypothesis or current goal. High priority is given to information that has specific properties or attributes (e.g. intensity or temporal recency). Knowledge of how cognitive processing interacts with incoming audio-visual information can greatly enhance the ability of the multi-modal model to predict human quality responses. The multi-modal model is directed towards developing a mathematical account of the sensory and perceptual properties of the stimulus and relating these properties to human opinions of quality. The advantage of understanding the cognitive processes involved in the human evaluation process is that the mathematics behind the multi-modal model can be fine-tuned to focus on the critical attributes of the stimulus,

thereby reducing much of the redundant information present in the sensory and perceptual data. Therefore in-built cognitive foundations can operate as a guide to the sensory/perceptual components within the multi-modal model in order to produce a more accurate and robust prediction of subjective quality opinions.

2.3 *Memory and quality evaluations*

This section takes a practical look at the relationship between memory and quality evaluations. In any sequence of events, some items are more likely to be recalled than others. For example, it is known that information presented at the beginning (primacy) or end (recency) of a sequence is more accessible to memory recall than middle-item information. Apart from the temporal properties of sequentially presented information, audio-video clips contain different types of audio-video content. Of course, some content may be very memorable and it is an important question to understand the role content memory has on quality evaluations. Are quality evaluations more severe if the poorest quality affects a highly memorable part of the clip compared to errors occurring in low memorable parts? Is there some relationship between a subject's ability to recall content and ratings of quality (i.e. is the relationship between memory and quality direct or indirect)?

An initial experiment was run to examine whether using memory recall as a second task in addition to the quality evaluation task influenced subjects' opinions of audio-video quality. Task effects are likely to have important implications for our understanding and modelling of subjective quality responses. For example, the traditional quality method simply asks subjects to view an audio-video clip at the end of which a quality rating is requested. This situation can be defined as passive — the subject does not have any real involvement with the audio-video presentation. How does this equate to the home viewing environment? In both viewing television and interacting with a computer (e.g. videoconference) the user is interested in what is being presented to them. The viewing position is now much more active (i.e. the information has some meaning). Thus, it is critical to know if a subject's perception of quality is different when quality is evaluated under passive and active viewing conditions.

The experiment presented seven different audio-video clips to subjects. At the end of each clip subjects were asked to provide a quality evaluation. Two groups of 12 subjects participated in the study. One group of subjects was asked to give a quality evaluation for the first six trials and at the end of the seventh trial both a quality rating as well as a surprise content recall task were requested. The second group of subjects was required to provide both a quality rating and recall the audio-video content for every trial. The results of the experiment showed that quality ratings for the

quality-only group were the same as ratings from the quality and recall group (see Fig 2). Therefore, the results of this study suggest that quality ratings are independent of whether subjects adopt a passive or active viewing/listening position towards the test material. Future tests in which subjects are asked to physically track some aspect of the test stimuli (e.g. as in a multimedia game environment) are necessary to extend the present results into situations where the subjects actively engage with the test stimulus.

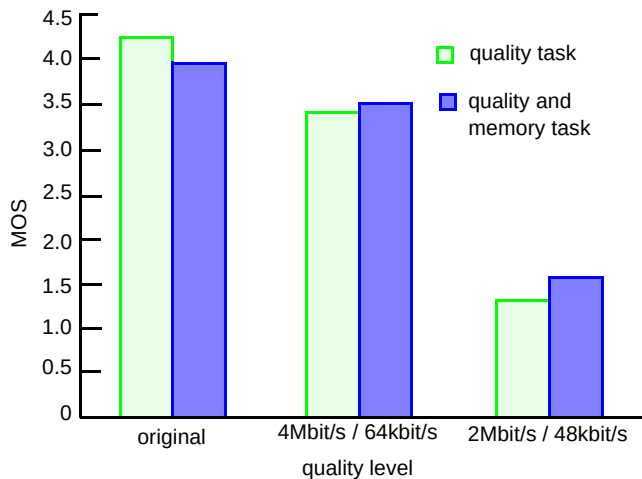


Fig 2 Mean opinion scores for both the quality-only task group and the quality and memory task group.

2.4 Task influence

As has already been stated, 'task' plays an important role in perceptions of multi-modal systems, and hence the model will need to have knowledge of this. The level of granularity into which task needs to be broken down for the purposes of modelling will be an important area of investigation. Can task be categorised simply at a very coarse level (such as passive watching, one-to-one conferencing, one to many remote lecturing, etc), or at a very fine level (e.g. at this moment in time the user was using the video link to examine a document closely)? In considering these factors, the practicalities of operating a predictive model need to be considered. The model will give predictions for the average user, not a given individual, and it is unlikely that every user for any given task will be paying attention to the same thing. Hence, it is likely that the definition of task for the model will tend to be more towards the coarse end of the spectrum. In collecting information to build the model, however, it may well be necessary to investigate task at a finer level of detail in order to gain an implicit understanding of how task interacts with perception.

Another important driver for the task aspect is business needs. This will serve mostly to constrain the potential task scenarios to those that are, or are likely to be, of interest to a global communications company such as BT.

3. Cross-modal interaction

In order to model multi-modal perception it is necessary to investigate a variety of cross-modal effects. This section introduces some key audio-visual cross-modal effects and describes example experimental data for model development.

3.1 Synchronisation perception

Synchronisation tolerances have been an essential consideration in television broadcasting for many years. However, for emerging telepresence technologies, where synchronisation must be dynamically controlled, it is necessary to describe synchronisation requirements in more detail. Audio/video synchronisation error detection is dependent on the task undertaken, the nature of the stimulus (content) and whether the error results in the audio leading or lagging the video [18]. These factors have been investigated in subjective experiments in order to provide data for the synchronisation component of the multisensory model. Data on synchronisation error detection is also of immediate use to the developers of videoconferencing and virtual-conferencing [19] systems so that optimal buffer handling can be designed.

A series of experimental investigations at BTL have been undertaken in order to provide data on the extent to which synchronisation detection varies with audio lead and lag, and the type of audio-visual content. The results presented in this paper were obtained by asking subjects to behave as 'synchronisation error spotters'. Subjects responded positively (say 'yes') if they perceived a synchronisation error and negatively (say 'no') if they did not. A description of this type of experiment has been published elsewhere [20].

Temporal asymmetry

Figure 3 shows the probability of detecting a synchronisation error averaged across all stimulus types. The underlying feature of the temporal asymmetry shown (where synchronisation errors are more noticeable when the audio cue leads the visual cue) is in many ways unsurprising since we are used to receiving audio cues later than visual cues. This is our experience in the natural world since the associated physical signals travel at radically different speeds.

The general form of the results reflects the recommended synchronisation thresholds given in ITU-T J.100 [18], i.e. 20 ms for audio lead and 40 ms for audio lag. This recommendation provides a fixed figure for all content types and is intended to ensure that synchronisation errors remain imperceptible. This is clearly the correct approach for the specification of broadcast systems.

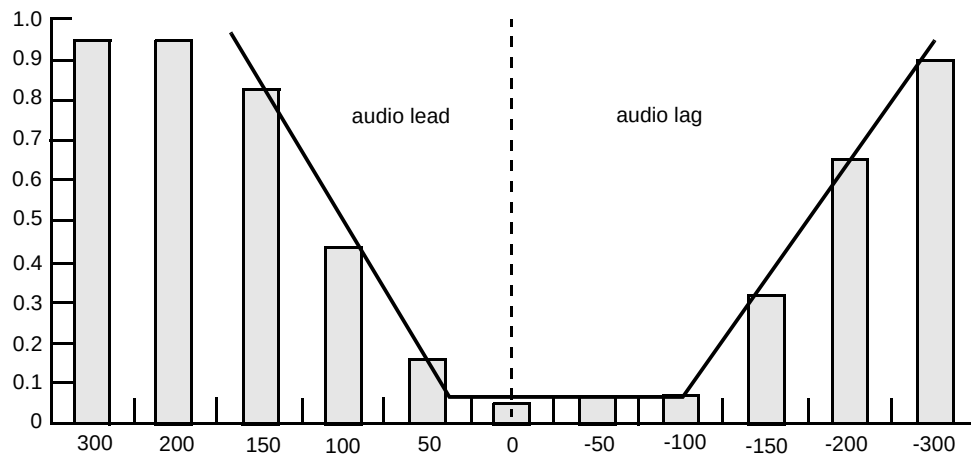


Fig 3 Synchronisation error detection (averaged for content) — audio lead, ms, versus probability of detection.

Content dependency

It is reasonable to assume that an observer's ability to detect asynchrony between audio and video cues will depend on the nature of those cues. However, to determine the relationship between the properties of audio/visual cues and the detection of synchronisation errors, experiments have been conducted with a number of different stimulus types.

The results presented in Figs 4 and 5 refer to three types of content which provide stimuli with various properties:

- bouncing pen, Figs 4a, 5a — 'short' visual cue,
- axe, Figs 4b, 5b — 'long' (persistent) visual cue,
- talking head, Figs 4c, 5c — visual speech cues.

It was hypothesised that synchronisation error detection would be maximum for the long visual cue, and that the short visual cue would be similar to visual speech cues. This hypothesis was based on the observation that there are short visual cues available within visual speech. It was acknowledged that higher level perceptual processes might also have an influence. Figure 6 shows the results for the three stimulus types on a single graph for ease of comparison.

The key features of these results are:

- the general trend in error detection asymmetry is apparent for all stimulus types,
- the duration/distinctness of the 'axe' stimulus has resulted in greater probability of detection than for the shorter 'bouncing pen' stimulus,

- error detection for the speech stimulus is consistent with the other stimuli when the audio lags the video, but greater than either the 'axe' or 'bouncing pen' stimuli when the audio leads the video.

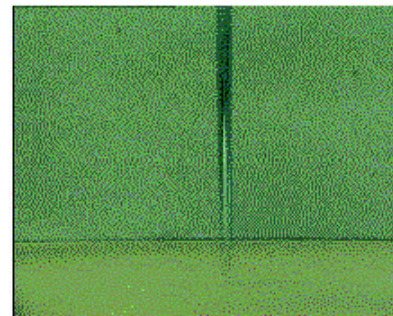


Fig 4a Bouncing pen stimulus.



Fig 4b Axe stimulus.



Fig 4c Talking head.

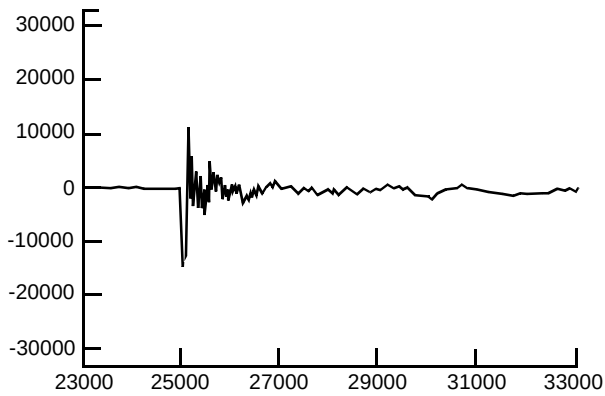


Fig 5a Bouncing pen — audio stimulus.

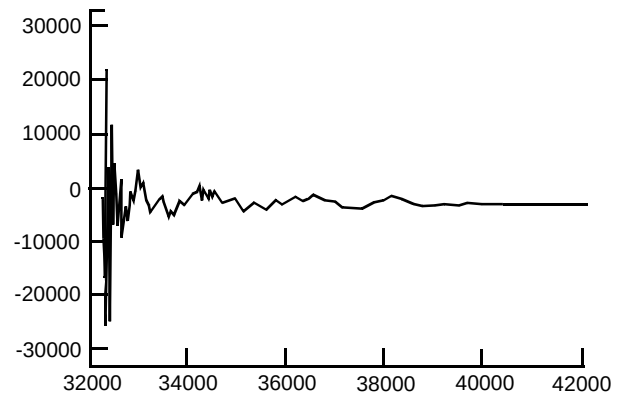


Fig 5b Axe — audio stimulus.

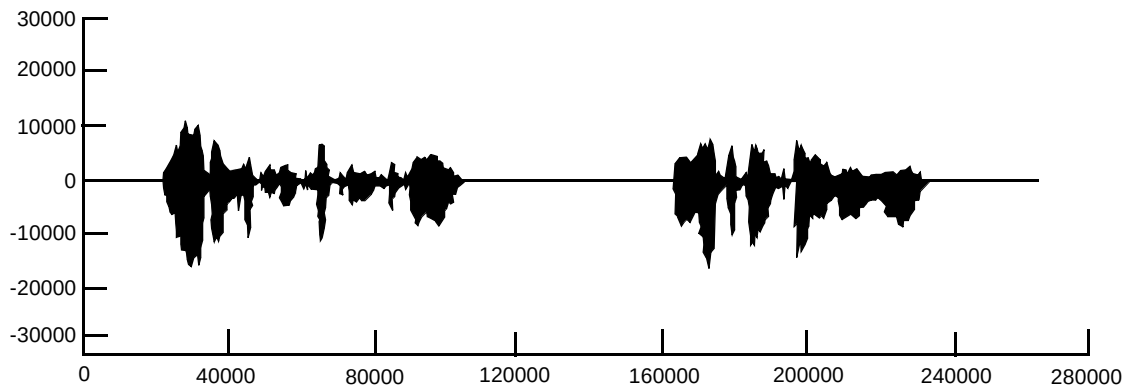


Fig 5c Talking head — audio stimulus.

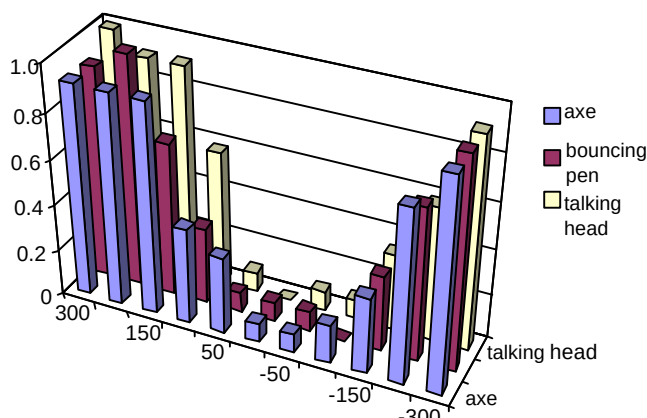


Fig 6 Synchronisation delay versus probability of detection for different stimulus types.

Suggestion for mathematical model

Experimental data describing content dependency and subjectivity of synchronisation errors can be used as the basis of a synchronisation perceptual model. This model will combine objective descriptions of audio and video events with rules concerning synchronisation error perception to provide an objective prediction of perceived error. The main parts of the synchronisation perceptual model are shown in Fig 7.

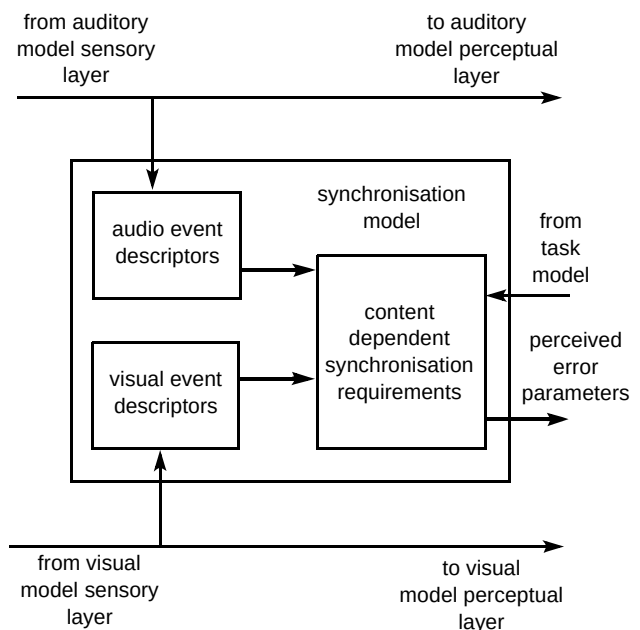


Fig 7 Synchronisation perceptual model.

In the prototype model the objective parameters describing the audio and video components of the signals are:

- audio descriptors — sliding 20 ms RMS energy, signal peak and decay parameter,

- video descriptors — motion vectors [21] and a persistence parameter describing the subjective importance and its decay over time.

Such a model fits within the multi-modal model framework proposed in Hollier and Voelcker [10] and shown in Fig 1. The model is also potentially useful as a stand-alone component for the dynamic optimisation of audio-video synchronisation

3.2 Cross-modal influences on quality perception

Quality perception and system optimisation

The assessment of quality in any multi-modal system is a time-varying multi-dimensional activity — sensory interaction changes from moment to moment requiring synchronous modelling. The two most common modalities encountered in communications systems are the auditory and visual senses. Information on cross-modal interaction in relation to perceived quality would be invaluable in a system designed to measure the quality of service (QoS) in multi-modal delivery systems, whereas individual assessment of the individual perceived modality qualities may miss some significant interaction-related problems (such as synchronisation or more subtle effects as discovered by the experiments described below). Knowledge gained could form the basis of a multi-modal codec where, say, the bit rate was varied between audio and video to give a specified QoS. Applications of the model are discussed in section 5.

The perceived quality of any system is clearly content-dependent — if one is reading text from a screen, the quality of the audio has little significance and, if one is casually chatting, the quality of the video is of less importance than that of the audio. When designing experiments with the intention of populating key areas of the multi-modal model, it is important to consider the task undertaken and to keep them representative. For the two experiments described in the next sub-section, the task was classified as ‘passive viewing’ — the subject sat and viewed a mixed audio-visual presentation giving a subjective quality rating at the end of each clip.

Subjective tests

A pair of subjective tests [22, 23] was designed to investigate the prominence of cross-modal interaction [10, 24 — 27] when assessing quality of an audio-visual system. Both experiments took a series of audio-visual sequences and degraded the individual audio and video quality [28, 29], producing a range of quality conditions, from perfect to barely intelligible. The subjects were each presented with a random selection of conditions and asked to give an opinion selected from a discrete five-point scale — *excellent*, *good*, *fair*, *poor* or *bad* [29]. By averaging the scores from each

subject a mean opinion score (MOS) was obtained for each condition.

The first of the two experiments used an animated fly-through with a narration and the second experiment used talking-head material (such as a newsreader). It was hypothesised that there exists a difference between the subjective quality perception of human faces and other objects — this was based on the fact that human perception is highly optimised for person-to-person communication (for example the auditory system is well matched to the frequencies found in human speech).

Figure 8 shows the results from the first (animation and narration) experiment and it can be seen that there is a small, statistically non-significant, change in perceived audio quality with a change in the video quality.

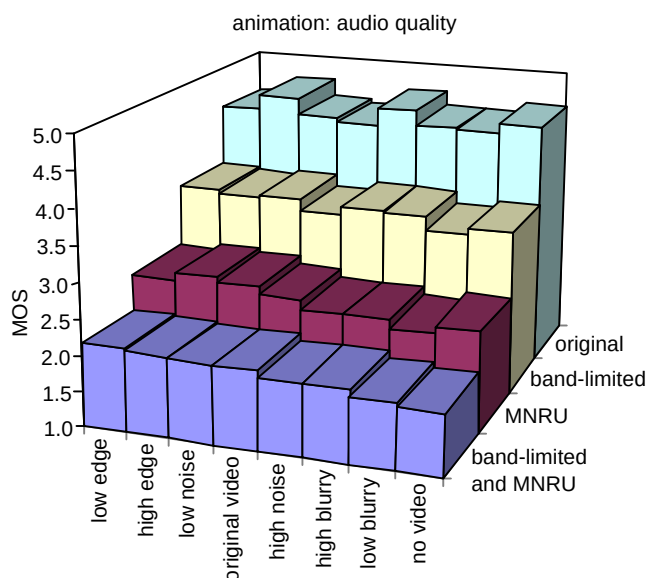


Fig 8 Animation experiment — MOS scores for audio over a range of eight levels of video degradation and four levels of audio degradation.

Figures 9 and 10 show the results from the second experiment. These graphs clearly show the strong inter-modal dependency between auditory and visual modes. Figure 9 shows how perceived audio quality (for a specific audio condition) changes as the video quality deteriorates. For example, for the undegraded audio conditions, when the video quality is high the audio MOS is 4.8 and when the video quality is poor the audio MOS is 2.3, even though the actual audio used is unchanged. The change in MOS is statistically significant, being equivalent to a change in opinion from *excellent* to *fair*.

As hypothesised, the cross-modal interaction is greater where the material comprises talking-head content. This was reinforced by anecdotal experimental evidence where subjects, upon completing the subjective test, complained of

visual distortion in the area of the eyes and mouth of a face — other distortion was nowhere near as disturbing.

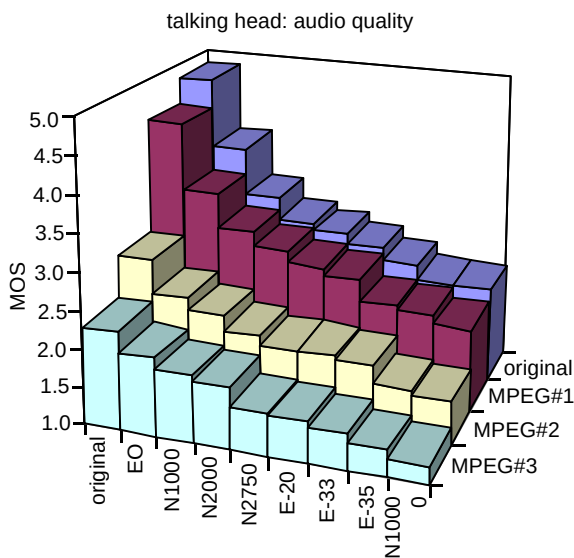


Fig 9 Talking-head experiment — MOS scores for audio quality over a range of nine levels of video degradation and four levels of audio degradation.

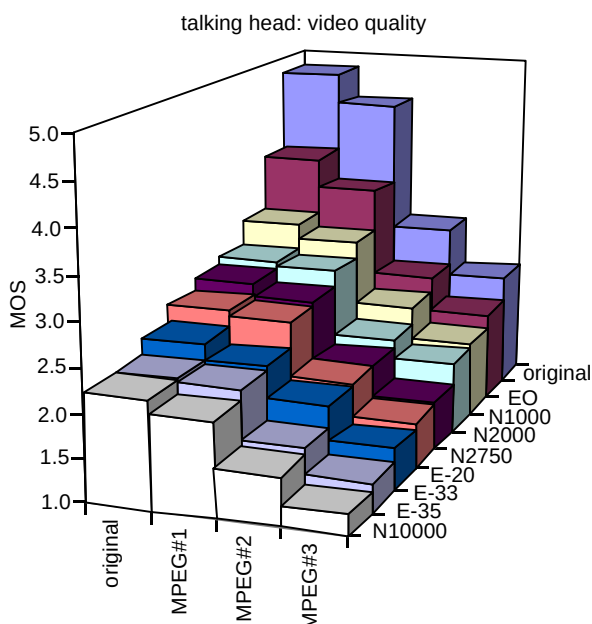


Fig 10 Talking-head experiment — MOS scores for video quality over a range of nine levels of video degradation and four levels of audio degradation.

The above findings are of interest when designing human/machine interfaces for communication purposes. Due to the increased sensitivity of error discrimination in human faces it is proposed that greater coding effort be given to the encoding of features such as eyes and lips. The overall effect would be that of a system (such as a video-conferencing system) with a greater perceived quality. However, it is important to note that increasing the resolution of lip movements is likely to affect the ability of

users to notice synchronisation anomalies between the audio and visual streams.

4. Next generation testing

In order to build models of multi-modal perception, a large amount of data needs to be collected reflecting those aspects of multi-modal systems that the models should predict and measure, and in which we are interested. As the models are to predict the perceptions of users of multi-modal systems, ideally, subjective quality responses would be gathered from people actually in their natural environments. However, to retain sufficient control in order to obtain representative and repeatable results, some compromise must be made between the artificiality of controlled test environments and the experimenter's level of control over the experiment. Therefore a flexible, yet controlled, environment is needed in which to conduct such experiments in order to immerse people in realistic environments where they can actually feel that they are performing tasks as they would in real life, and are not in what would be considered to be laboratory conditions. This is the rationale behind the construction of the new Perception Laboratory at BT Laboratories, which is a facility dedicated to the assessment of a wide range of multimedia systems.

With this need for more realistic assessments, new methodologies will need to be devised as those traditionally employed to investigate subjective quality perception rely on very strict laboratory conditions. The nature of these new methodologies will be dependent on the outcome of the investigations into task dependency and the practicality of the various possibilities for data collection.

Firstly, the issue of task dependency must be addressed — as has been discussed earlier, when considering people's opinions of multi-modal systems, task is known to play a crucial role, and can have a strong influence on the opinions of the users. In collecting data for building models of multi-modal perception, it is therefore imperative that these task effects are accounted for, and as such are reflected in data collection experiments. As it becomes clearer (through both theoretical and experimental work) what level of granularity is required to define tasks, specific experimental tasks will be devised to represent particular task groups (which may include passive viewing, one-to-one information exchange, one-to-many lecture style, etc).

Secondly, on data collection methods, it is necessary to collect the right data in the right way. Traditional methods use a number of different rating scales (e.g. comparison scales, discrete and continuous category scales) and subjects are normally informed that their task is one of quality evaluation and are requested to refer to a set of predefined quality values (e.g. *good*, *fair*, *bad*) to represent their opinion of quality. For multi-modal modelling work, this

will certainly be an important method, but other methods will also be needed, such as continuous non-intrusive measurements, post-test questionnaires and/or focus groups. The reasons behind this need are two-fold — firstly, the dictates of the work on the taxonomy of task may well require that we need to collect continuous data to be able to model the relevant effects, and, secondly, the tasks themselves may necessitate the collection of intermediate data due simply to their length and complexity. This poses significant problems, as it is likely that subjects will be given a task in which the primary purpose is to be representative of typical usage rather than to concentrate on quality assessment — to interrupt such a task with possibly repeated questions, on say perceived quality, would by its very nature be distracting from the task itself and hence inappropriate. Therefore alternative methods of collecting data need to be realised, which may be related to physiological data, task performance, and user behaviour. In considering these techniques it will be important to consider that the data collected must be related to the quantities the model is to predict. The use of post-test questionnaires and/or focus groups will be essential in obtaining information that it may not be possible to collect formally as the experiment progresses due to factors such as non-intrusiveness and time constraints.

5. Future applications of multi-modal perceptual models

This section discusses how the proposed multi-modal model can be used in real-world applications.

5.1 System assessment

The first and most obvious application for the multi-modal model is that of a quality assessment tool. Much of the data gathered so far for use in designing the model is based on quality assessment or synchronisation detection.

By running a multi-modal model it should be possible to give an accurate prediction of perceived quality of the overall system or of one mode of a system (but considering the influences of other modes). For example, if the model predicted the picture quality of a videoconferencing system, it would also take into account the audio quality just as human test subjects do (as shown by the experiments in section 3.2).

As discussed in section 2.4, a major influence on subjective quality prediction is the task being undertaken. Any system designed to predict subjective quality must consider task and other influencing factors such as environment (lighting levels, background noise, etc) and possibly the expected emotional state of a subject.

A system, which can accurately predict the subjective quality of a system (single mode or overall quality), would be an essential tool for a service provider as it can provide a

real-time QoS measure. To obtain such information from actual subjects is difficult, expensive and time-consuming. The auditory model developed at BTL [30] (which forms the base of the auditory component of the multi-modal model) has been used in a real-world environment to give real-time QoS prediction of telephony systems and has proved the usefulness of such real-time measurement systems.

5.2 Delivery optimisation

Systems that incorporate models of human perception have been in use for many years — a well-known example is the MPEG audio codec [31]. The codec includes a model of auditory perception to determine which parts of the audio are audible and which are not; it then reduces the data bit rate by only encoding the perceptually significant components. The coding system is lossy (it is not possible to recreate an exact copy of the original signal). Considering the simplicity of the model, compared to that described by ISO [31], the coded signal can be indistinguishable from the original and the data reduction considerable — as the bit rate is reduced the coded signal sounds less like the original. A corresponding technique for data reduction of image files has been defined by the JPEG standard [32]. Current systems work by separately coding the audio and video components of the source material; while this does indeed provide a considerable data reduction, it takes no account of the cross-modal interaction.

The multi-modal model presented here could be used as the core of a multi-modal codec that, for a given bit rate, adjusts the data reduction of the individual modalities to give the optimum overall perceived quality. Such a system would ensure that at any time the perceptually most significant components of the presentation are transmitted while a lower bit resolution could be used for the remaining material.

A second application of a delivery optimisation system would be one that adjusts the control interface of, say, a virtual world depending on a user's performance. For example, a system may reinforce visual cues with spatial audio cues if a user is considered to be struggling with the system — knowledge of modal interaction would help to provide a natural and unobtrusive intelligent interface.

5.3 Intelligent systems

The applications described above form the basis of intelligent systems — they either assume or are informed about external factors influencing the perception of a multi-modal information stream. External factors such as task, environmental conditions or emotional state impact upon one's perception of quality or usability of a system.

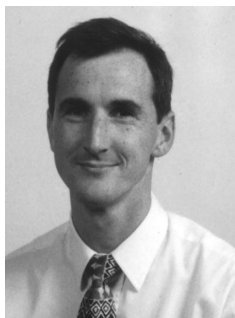
If a system is designed to adjust to the user it must do so extremely well otherwise it is better not to try to adjust at all. Consider the example of a computer which could detect emotional state. If a user, preparing a presentation, was very anxious because they had only ten minutes remaining, the computer might sense the anxiety and shut down to help them calm down — this would only exacerbate the situation making the user even more anxious. On the other hand an intelligent system which works well will remain unnoticed by the user and will help them to complete their task more comfortably and efficiently.

Acknowledgements

The authors would like to express their gratitude to colleagues at BT Laboratories and Jon May of Sheffield University who have contributed many comments and ideas to the development of multi-modal models.

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